

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 21.8

Audio Option

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1. INTRODUCTION

This chapter describes the installation of the audio module and hifi audio set.

The appendices in this chapter describe the audio functions and specifications of the audio module, music and patient headset.

The installation procedure of the headset is described in the installation manual that is delivered with the headset option.

One software key enables all music functions discussed in this chapter

The music option enables the CD-ROM drive of the host computer to be used for audio CD's and enables the use of locally purchased hifi audio equipment on the system.

The drawings manual section Operator Console describes the functional diagrams and the cable connections for the audio options.

2. INSTALLATION

2.1. INSTALLATION OF AUDIO MODULE AND HIFI AUDIO SET

1. The audio module is placed on the operator's console. Connect the audio module to BEA:X4. This is located on the back-side of the Backend Miscellaneous box on the Audio Switch board.
2. To install the hifi audio set or the CDROM drive as audio source, the following cables are shipped with the Gyroscan system:

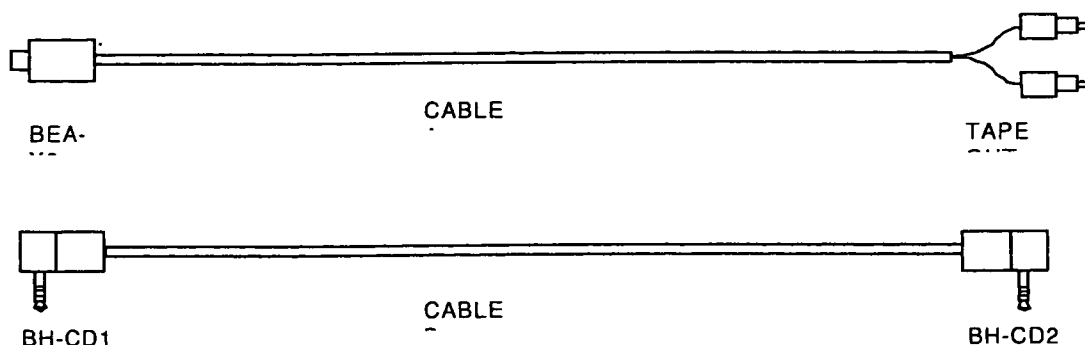


Figure 1

Cable A: AUDIO cable 4522 131 1895.

Connector: MiniDin 4p connector (BEA-X8) and 2x CINCH (TAPE-OUT)

Cable B: CD-ROM Cable 4522 131 2893., already mounted in the host console.

Connector: 2x stereo headset type (JACKET 3.5mm)

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3. If a hifi audio set is available, connect the audio cable (cable A) between the tape output connectors of the hifi set and BEA:X8.

If this Tape Out output is not available, then look for an other output:

- Use the Pre Amp output (if available) from the hifi audio set.
- Use the DCC output (if available) from the hifi audio set. However, the DCC player cannot be used as a recorder in this configuration.

Note that some audio amplifiers switch off the DCC output when the DCC is selected as input source. In that case, use another input to connect the DCC, for instance the AUX input.

NOTE

To avoid noise and hum on the audio module system, only connect above-mentioned cable A to the audio switch board if a hifi audio set is available. The loose cable end may pick-up noise signals.

4. Connect the power supply of the Audio set to the 230 V Mains Outlet in the Service Storage Cabinet (SSC). Extra mains connectors are delivered with the system for this purpose.
You have to cut the 'old' mains connector from the hifi audio set, and install the new connector.
If you need a 110V connection, a Mains Transformer has to be obtained locally. Specification: Input 230V 50/60Hz, Output 123-127V, Power < 200VA.

CAUTION

Use the spare mains outlets on the Mains Outlet in the SSC to avoid noise and hum in the audio system and to avoid electrical interference of the Gyroscan system.

5. Boot the computer. Login with user Gyrotool and install the software key for the patient audio option.

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2.2. FUNCTIONAL TEST OF AUDIO MODULE AND AUDIO SYSTEM

1. Check the functions of the audio module. The audio module functions can either be checked by TSW and by ASW.
 - Login with user system.
 - Select: configure hardware options
 - Select: configure CDROM as audio device.
 - Logout from user system and login with user TSW.
 - Enter: **TPATCOM**
 - Enter: **Run Tools Interactive**.
 - These tests check the complete audio chain.
 - Execute the tests. No errors are allowed.
2. Check the functions of the hifi audio system.
 - Login with user Gyroscan.
 - Press the Music On/off button on the audio module. It's LED must be ON to pass the music to the examination room.
 - To check the hifi audio system, follow the Gyroscan User's manual and/or the audio set's operator manual for the instructions.
3. Check the function of the CDROM drive as audio device. Select track 1, play, and check that the music over the ceiling speakers or headset in the examination room is present..

NOTE

The CDROM drive is default configured to load software. If you configured the CDROM drive as audio device, you cannot install software. To load software, re-configure the CDROM drive as SW load device.

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3. APPENDIX A: AUDIO FUNCTIONS

3.1. OVERVIEW OF AUDIO COMPONENTS

Refer to the table next page for an explanation of items mentioned below.

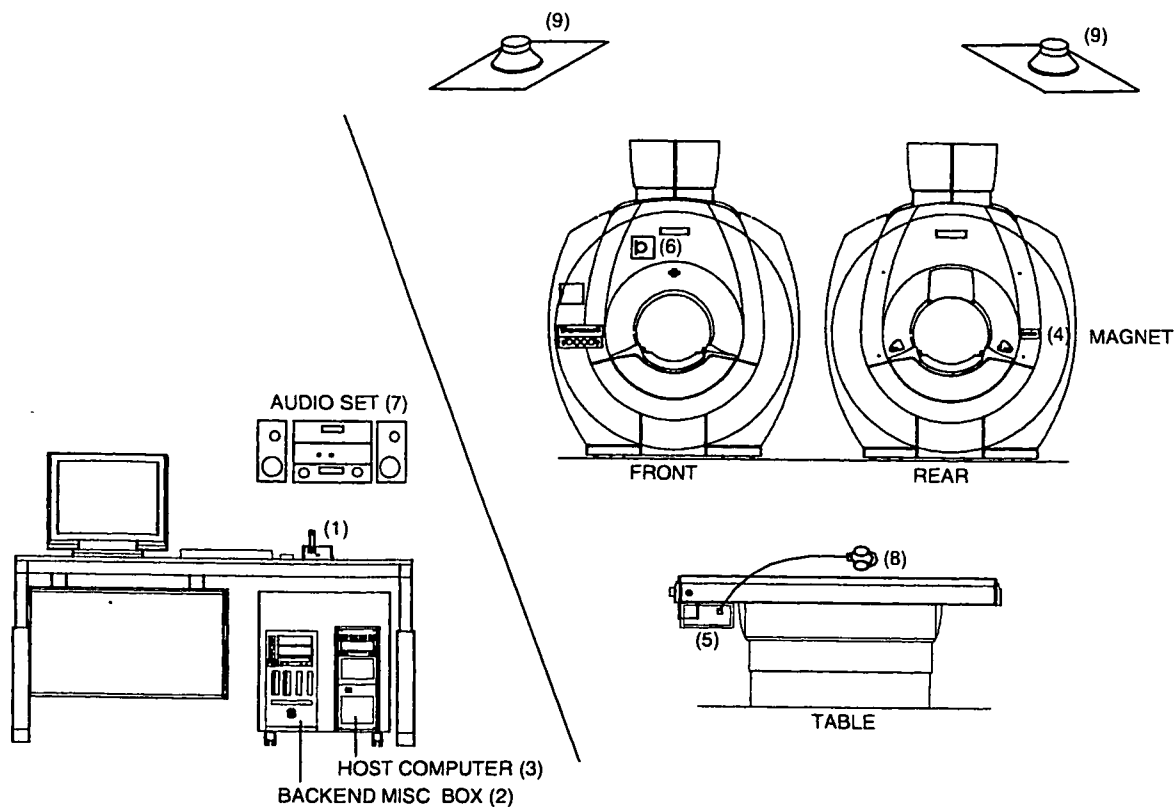


Figure 2 Components of audio system

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Number	Description of audio component
1	Audio Module
2	Expansion Console with built-in backend miscellaneous box.
3	Host Computer
4	PICUREAR_MT
5	PICU2KB
6	Microphone Box
7	Hifi audio set (optional)
8	Patient Headset (optional)
9	Ceiling speakers

1. Audio module.

The newly designed audio module (AM2) is used as intercom to communicate between the operator in the operator's room and the patient in the examination room. It contains a microphone, a loudspeaker, some LED's and buttons and two volume controls.

2. Backend Miscellaneous Box (BE).

This box contains the IIM, the M.O. disk drive, audio switch, RS232 board, and the backpanel for audio switch and RS232 board, two fans and the power supply. The backend misc. box is placed inside the host expansion console.

The audio switch board is routing the audio signals from various sources (CDROM, hifi audio set) to headset, audio module and ceiling speakers and contains the audio amplifiers.

The RS232 fibre board is an interface for eight serial links (RS232) of the Host computer. Besides communication to other system components, the RS232 board controls the communication between the host computer and the audio switch.

3. Host computer.

All audio functions are controlled by the software (patcom process) on the computer.

4. PICU2KB and PICUREAR_MT

5. The PICU2KB (a module inside the PICU Front) contains a Talk button to enable communication to the patient. The music volume can be controlled through the volume control on the PICU2KB.

The PICUREAR_MT (PICU Rear) contains a Talk button to enable communication to the patient.

The PICU2KB and PICUREAR_MT contain also buttons for scan control. PICUREAR_MT and PICU2KB are connected to the PICU which is mounted on the magnet. The PICU communicates to the host computer via a fibre link and the RS232 board.

6. Microphone

Inside the bore of the magnet (behind the cover) is a microphone. The microphone is used to talk to the operator. If the headset is present, the microphone can also be used to talk to the patient.

The microphone is connected to the PICU2BP.

7. Hifi audio set:

Option. See paragraph 4.3.2

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8. Patient headset

Option. The Patient Headset is used for passive noise reduction, for music and to communicate with the patient. The headset contains 3 tubes; one for each ear and one for the microphone. The patient headset is connected to the patient headset interface.

The patient headset interface contains two piezo speakers, the microphone for the headset and a detection switch to check if the tube assembly is correctly connected to the headset interface.

9. Ceiling speakers.

The ceiling speakers are used for the patient communication and for music.

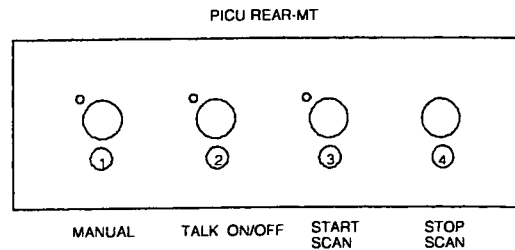
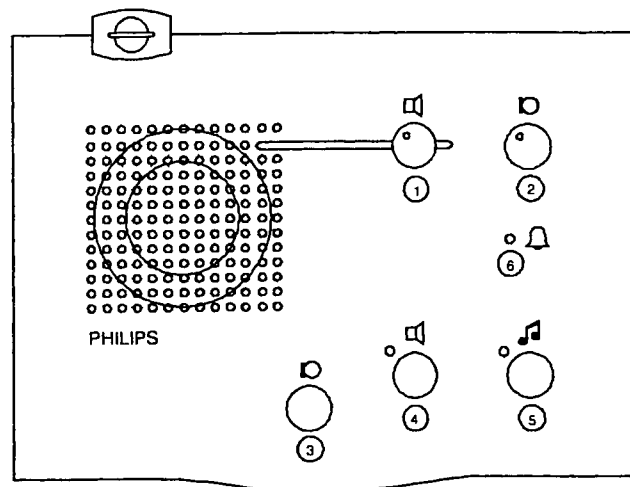
RF-feedthrough filters are part of the system filter box. You need special (non-spiking) connection cables for the ceiling speakers, which are delivered with the Gyroscan system.

3.2. OVERVIEW OF AUDIO MODULE FUNCTIONS

The following table shows the functions that can be controlled by the audio module and PICU's. The items in brackets () refer to the labels in the figures.

Function	Audio module	PICU2KB (front)	PICUREAR_MT (rear)
Music volume control. Adjust music volume to ceiling speakers or headset.	X (1)	x (2)	
Listen volume control. Adjust volume of patient's voice to the loudspeaker in the audio module.	x (1)		
Talk volume control. Adjust volume of operator's voice to the ceiling speakers or headset.	x (2)		
Talk button; press while talking.	x (3)		
Listen on/off. Enable/disable communication between operator and patient.	x (4)		
Music on/off. Enable/disable music to patient.	x (5)		
Talk on/off. In combination with patient headset.		x (1)	x (2)
Nurse call LED. The Nurse Call LED is blinking if the patient pressed the Nurse Call button for assistance.	x (6)		

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**Figure 3 PICUREAR_MT****Figure 4 Audio module**

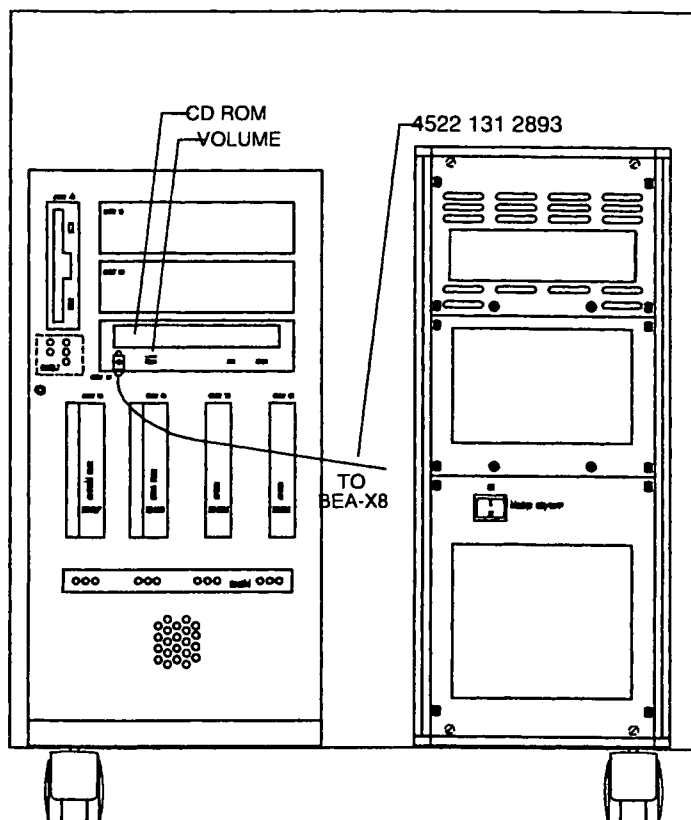
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3.3. OVERVIEW OF POSSIBLE MUSIC CONFIGURATIONS

3.3.1. CD-ROM DRIVE AS AUDIO CD-PLAYER

The CD-ROM drive is used for software loading, but can also be used for music (option). A software key is needed to configure the CD-ROM drive and to open a screen window for controlling the CD player. A factory-installed cable is connected to the headphone connector in front of the CD-ROM and to a connector in front of the host console. The volume on the CD-ROM must be set to maximum.

Figure 5



GR64 DRW

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3.3.2. HIFI AUDIO SET

The hifi audio set should be bought locally. It may contain any type of audio device such as CD, TAPE recorder, Cassette recorder or tuner. The loudspeakers can be used for music in the operator room.

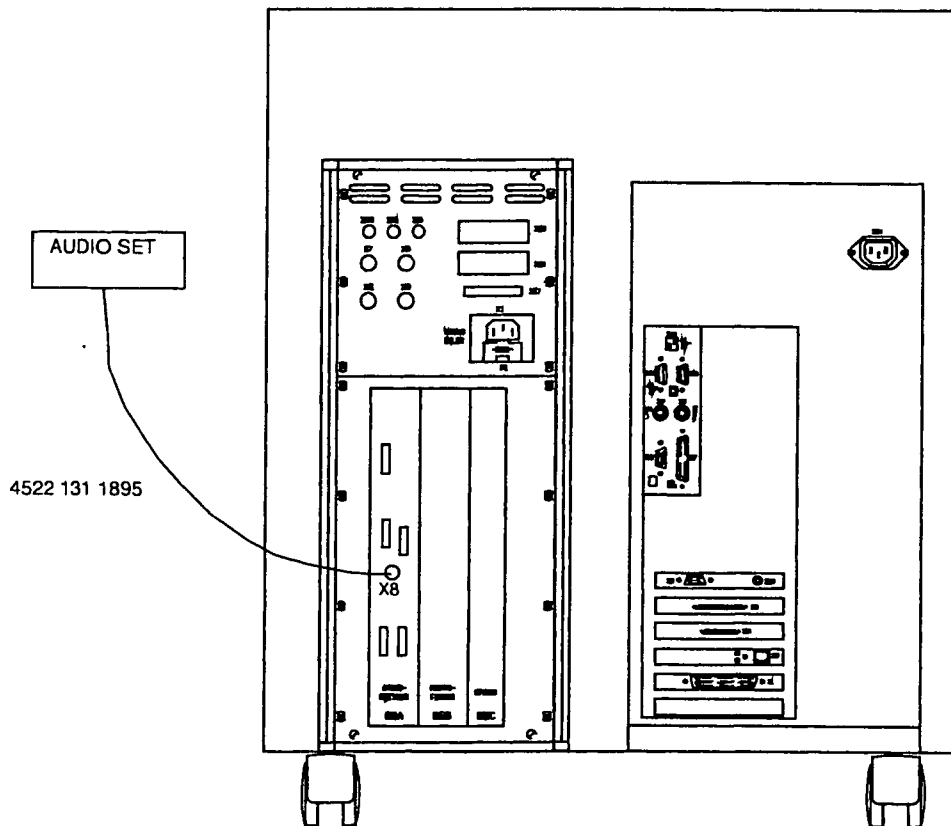
CAUTION

The loudspeaker boxes of the hifi set cannot be used in the examination room.

If the hifi audio set has an audio output (tape out), it can be connected to the audio switch board in the expansion console. Hence, the music from the hifi audio set can be heard in the examination room via the ceiling speakers or via the patient headset.

The RC-5 communication bus is not supported any more.

Figure 6 Hifi audioset connected to MR system.



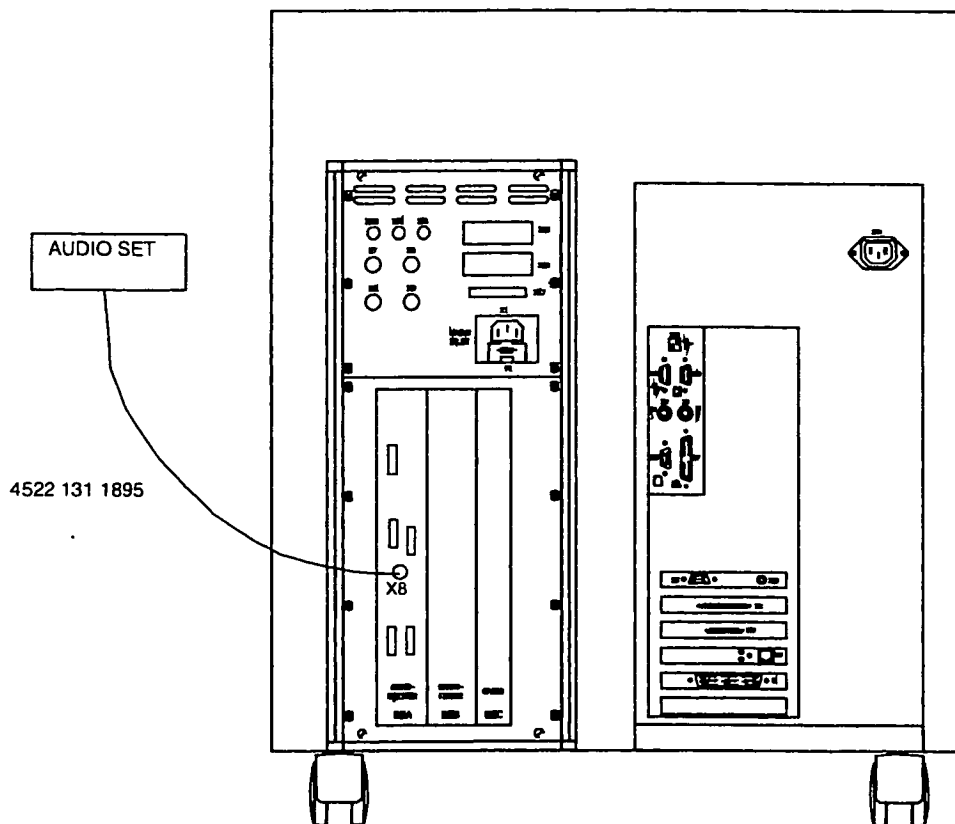
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3.3.3. SINGLE CD OR OTHER AUDIO COMPONENT

You can use a single audio component such as a CD player or any other audio device as music source. It's audio can be heard in the examination room via the ceiling speakers or via the patient headset. The control room will not have any audio from this audio component.

You can connect the audio output of this equipment, with cable BIA-X8 - TAPE OUT (Cinch plugs), to the audio switch board

Figure 7



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4. APPENDIX B: SPECIFICATIONS OF AUDIO EQUIPMENT

4.1. CEILING LOUDSPEAKER SPECIFICATIONS

Power : 9 Watt
Freq. response : 150 Hz to 16.000 Hz
Impedance : 4 Ohms

CAUTION

*Use the speaker connection cables that are delivered with the Gyroscan system.
Improper cables can cause image artifacts due to spiking.*

4.2. POWER AMPLIFIER IN AUDIO SWITCH

Audio Output : stereo, 2 x 9 W. R.M.S. at 4 Ohms.

Line input : 150 - 350 mV audio
Connector : 4p mini-DIN